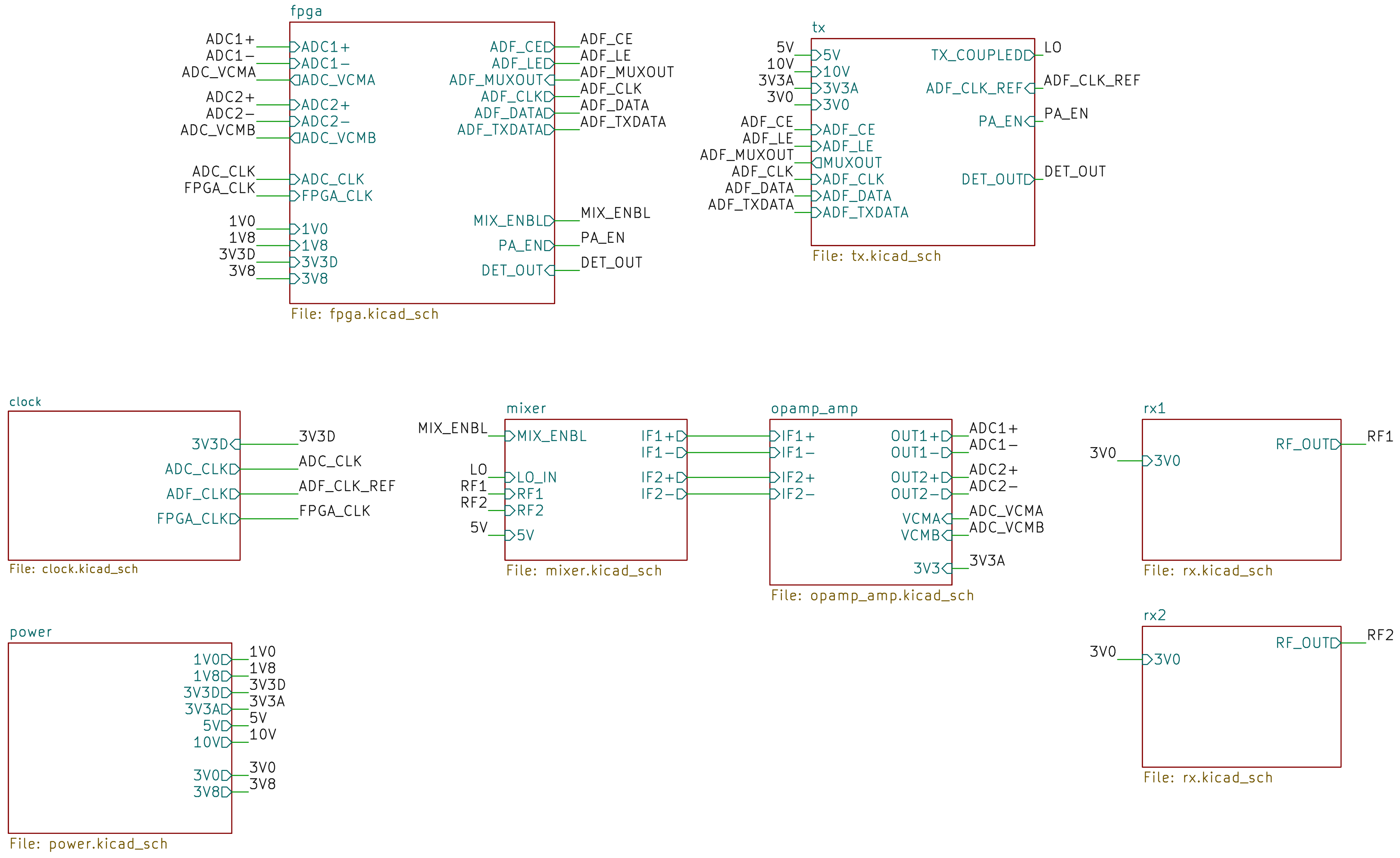
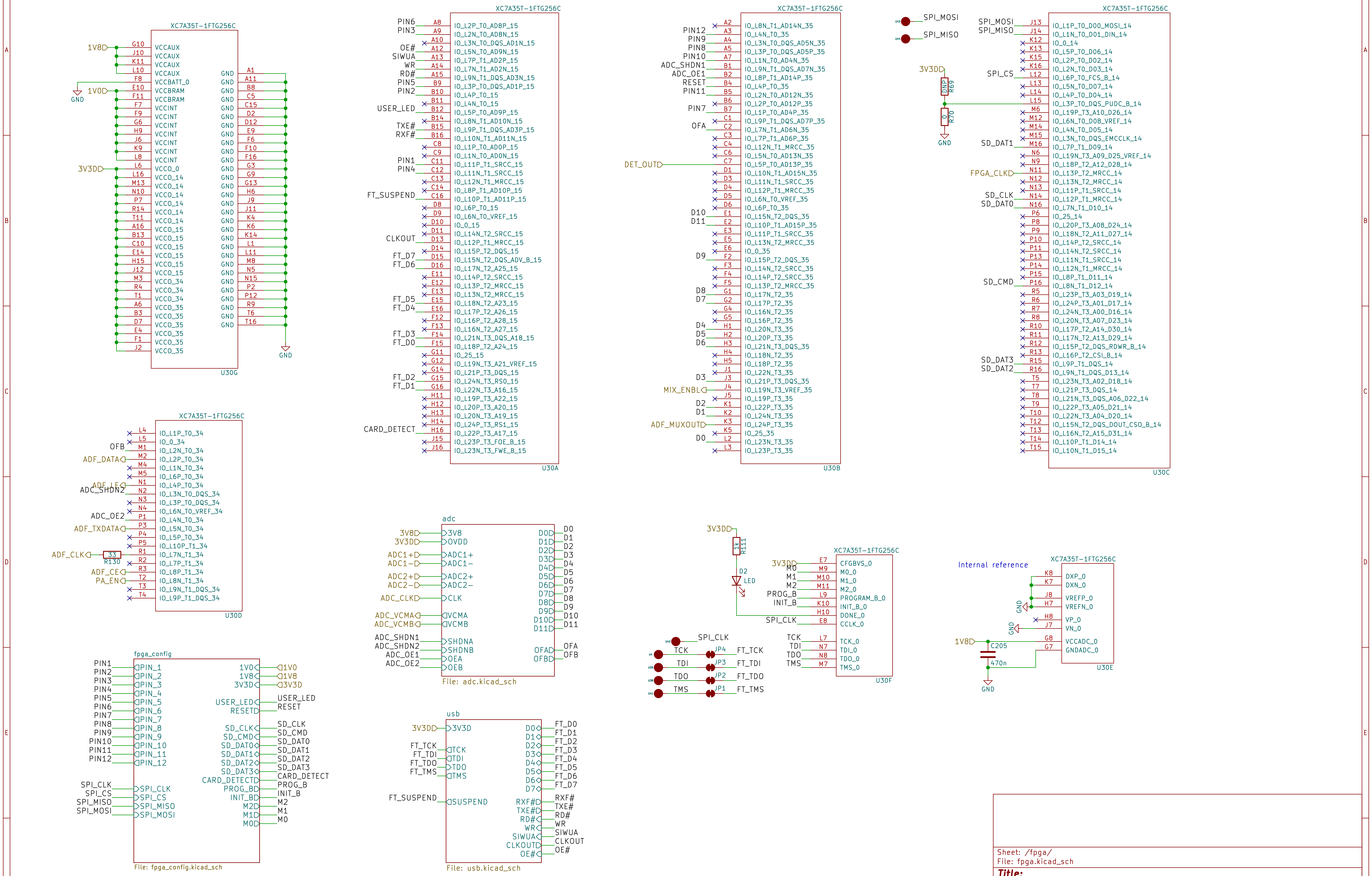


Sheet: /
File: fmcw3.kicad_sch

Title: FMCW3 by CENK KESKİN

Size: A4 Date:
KiCad E.D.A. 9.0.7

Rev:
Id: 1/12



Sheet: /fpga/
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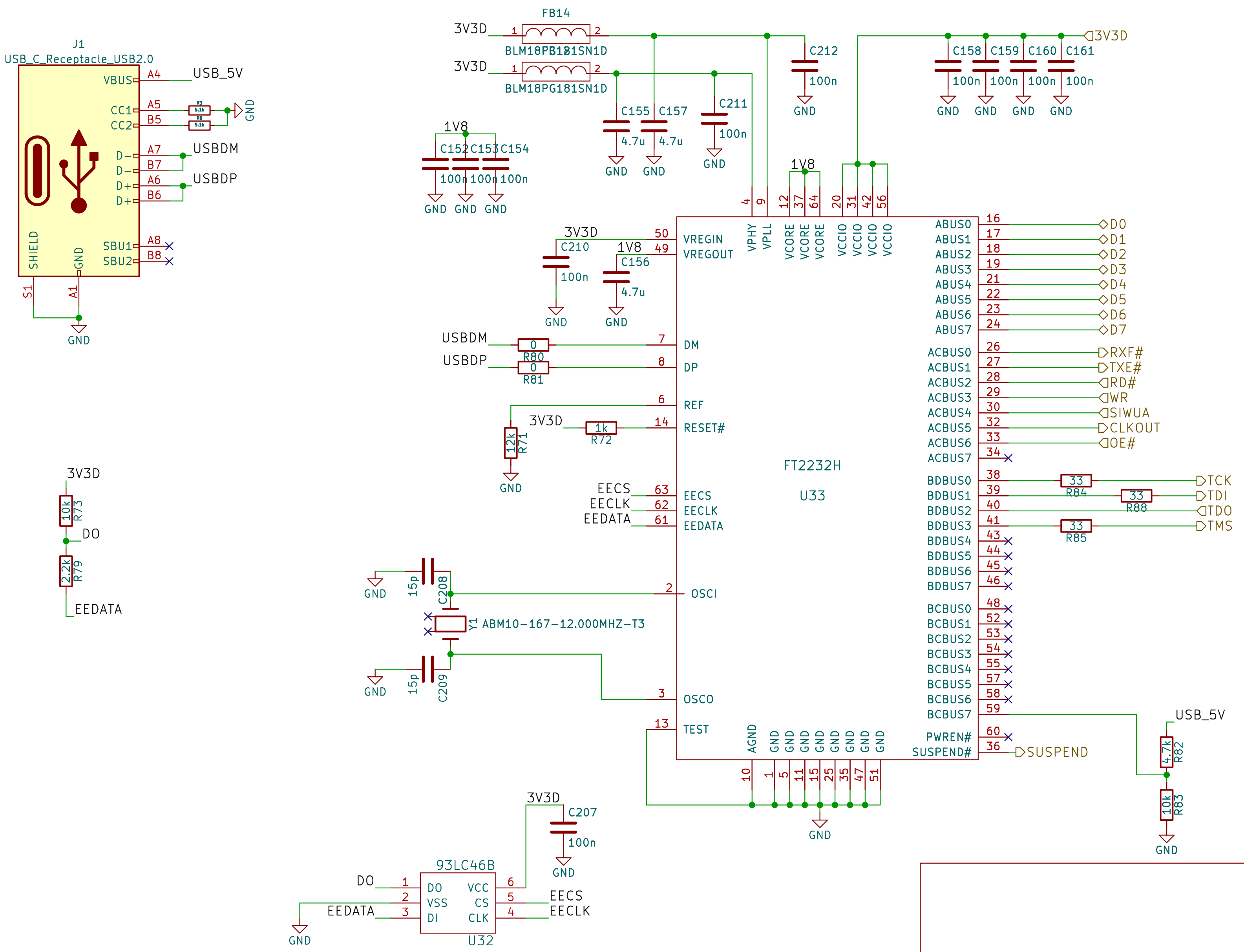
Size: A3	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 2/12



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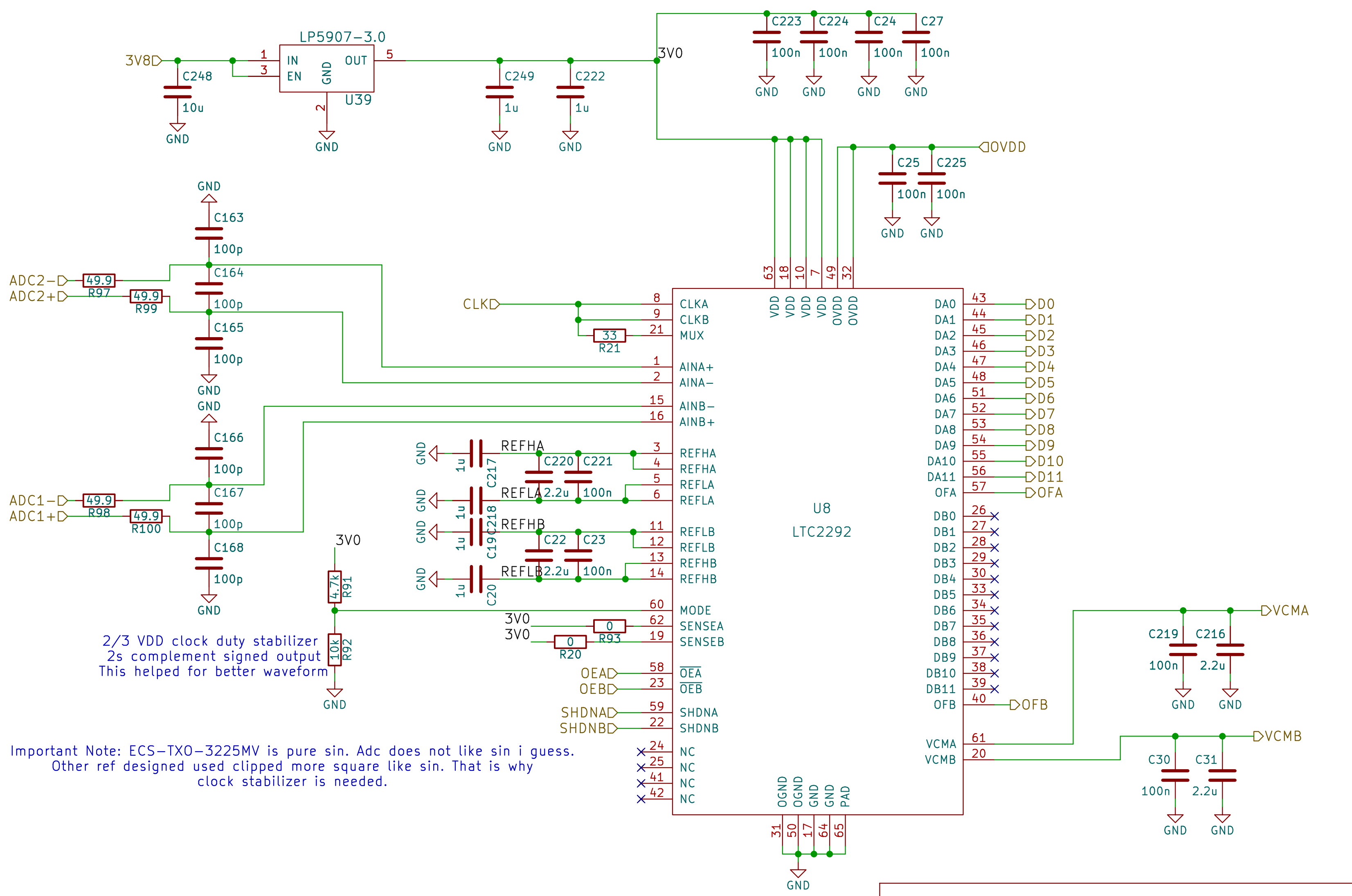
File: usb.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 3/12

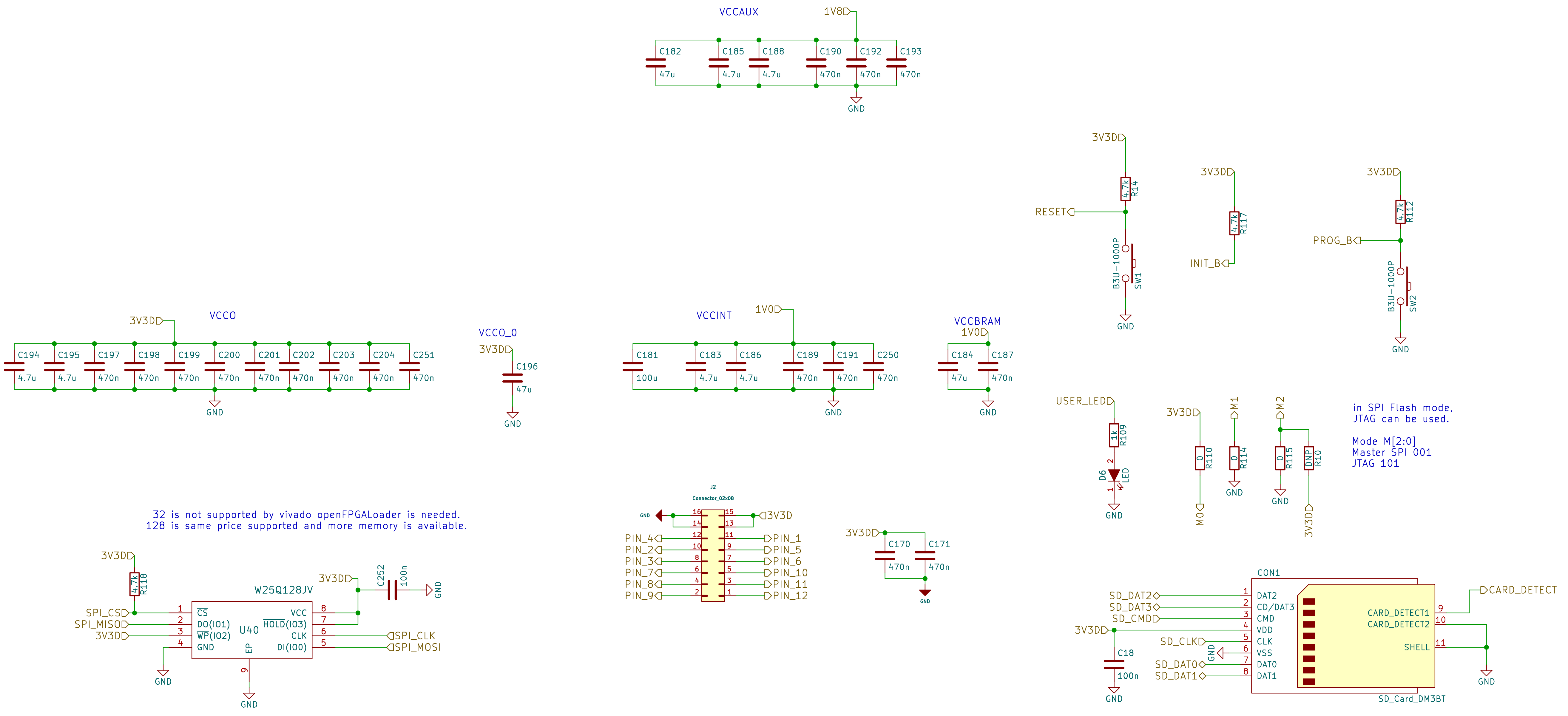


Sheet: /fpga/adc/
File: adc.kicad_sch

Title:

Size: A4 Date: Rev:

KiCad E.D.A. 9.0.7 Id: 4/12

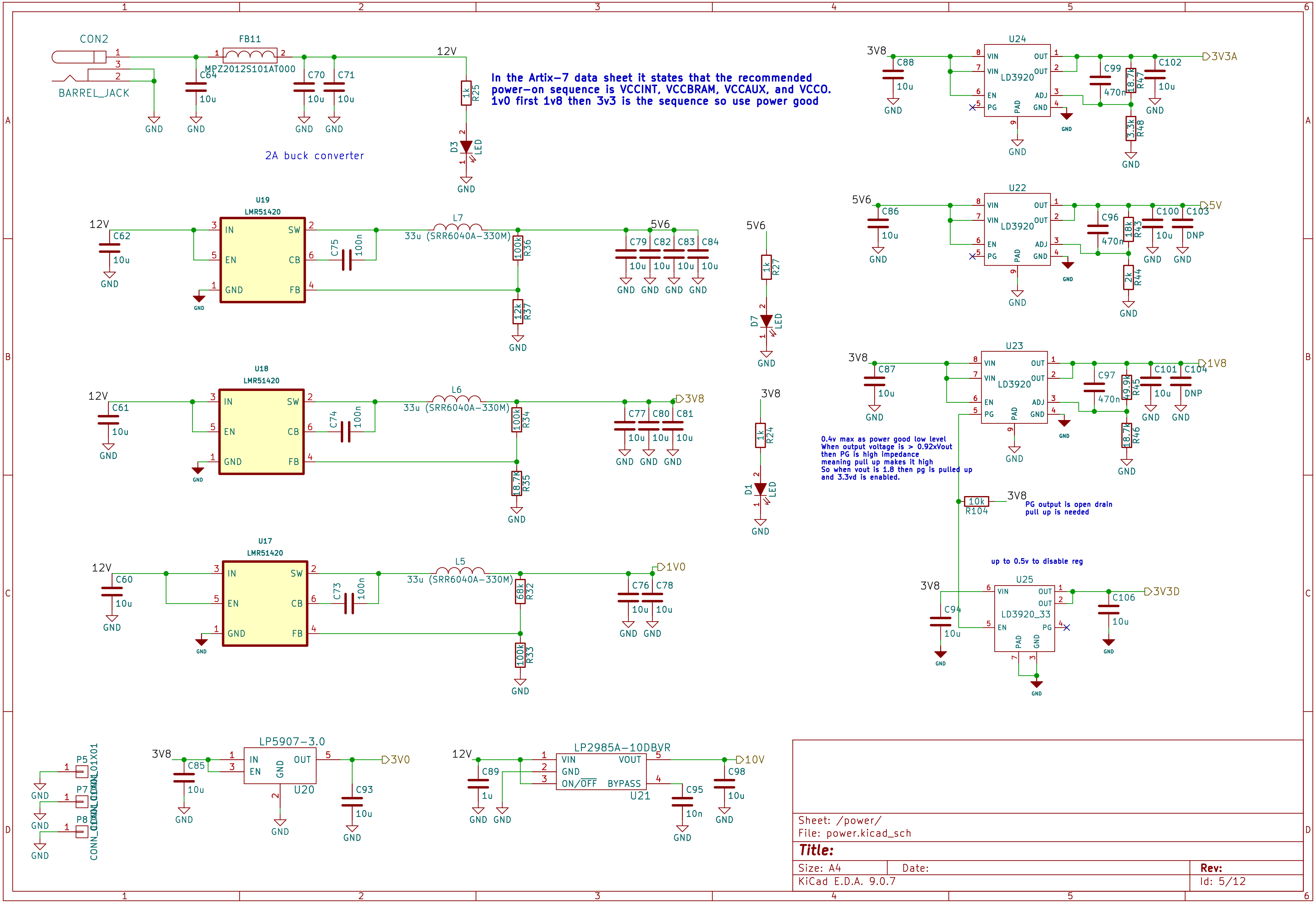


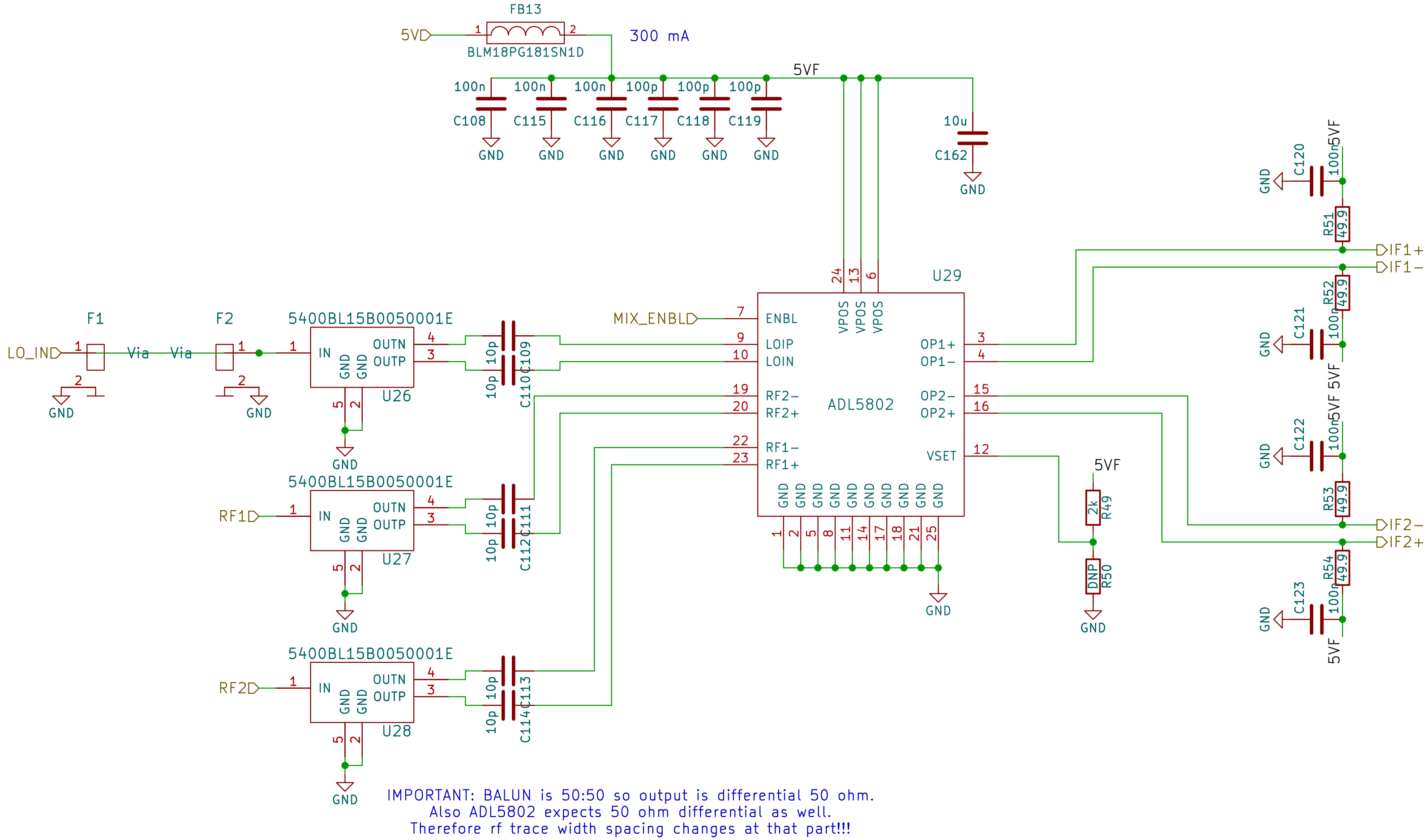
Sheet: /fpga/fpga_config/
File: fpga_config.kicad_sch

Title:

Size: A3 | Date: | Rev:

KiCad E.D.A. 9.0.7 | Id: 12/12

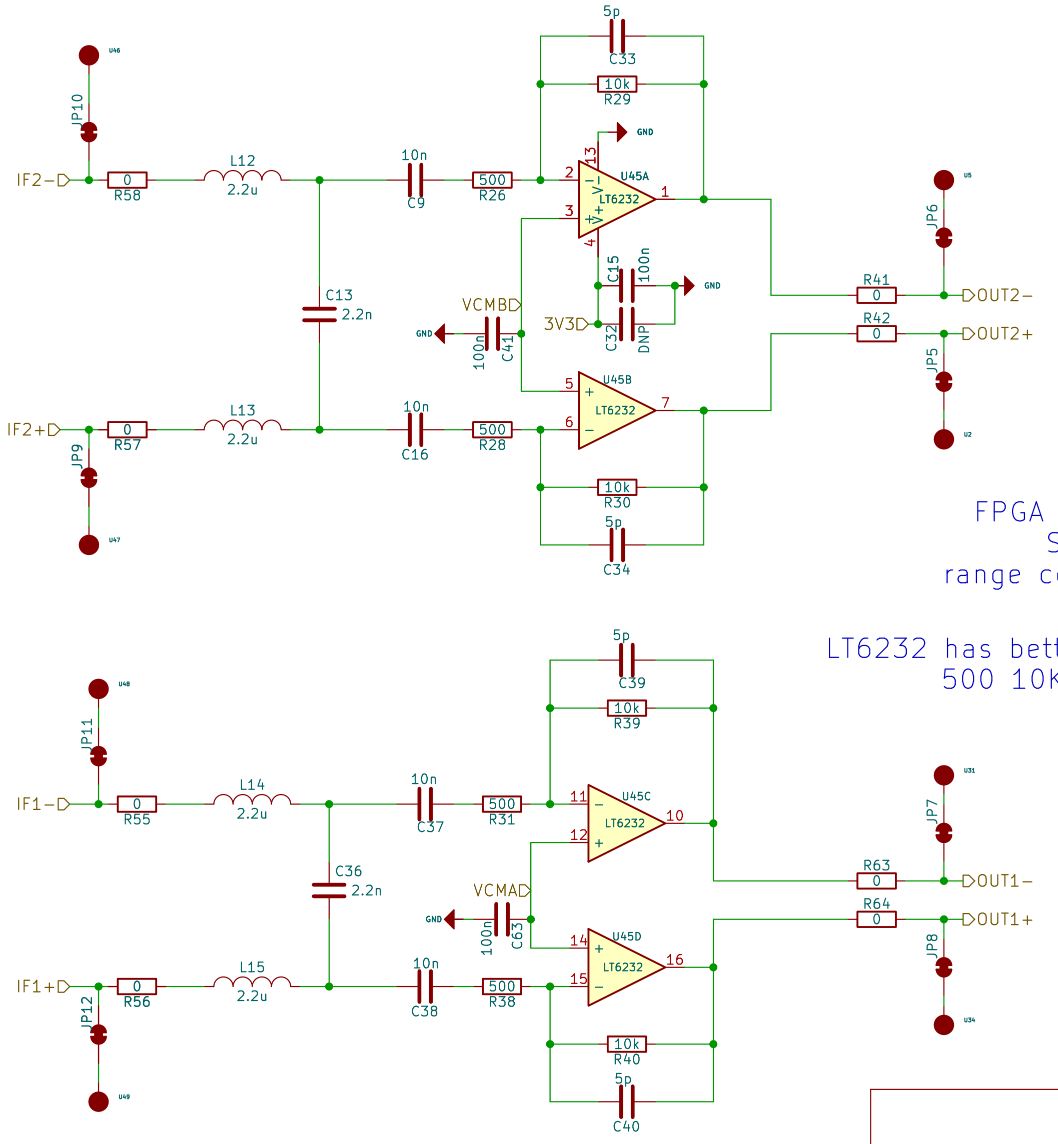

In the Artix-7 data sheet it states that the recommended
power-on sequence is VCCINT, VCCBRAM, VCCAUX, and VCCO.
1v0 first 1v8 then 3v3 is the sequence so use power good
2A buck converter
CON2
BARREL_JACK
FB11
MPZ2012S101AT000
12V
5V6
3V8
U19 LMR51420
U18 LMR51420
U17 LMR51420
L7 33u (SRR6040A-330M)
L6 33u (SRR6040A-330M)
L5 33u (SRR6040A-330M)
U24 LD3920
U22 LD3920
U23 LD3920
U25 LD3920_33
3V3A
5V
1V8
3V8
1V0
3V3D
3V0
10V
0.4v max as power good low level
When output voltage is > 0.92xVout
then PG is high impedance
meaning pull up makes it high
So when vout is 1.8 then pg is pulled up
and 3.3vd is enabled.
PG output is open drain
pull up is needed
up to 0.5v to disable reg
LP5907-3.0
U20
LP2985A-10DBVR
U21
CONN_01X01
Sheet: /power/
File: power.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 9.0.7
Id: 5/12



Sheet: /mixer/
File: mixer.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 6/12



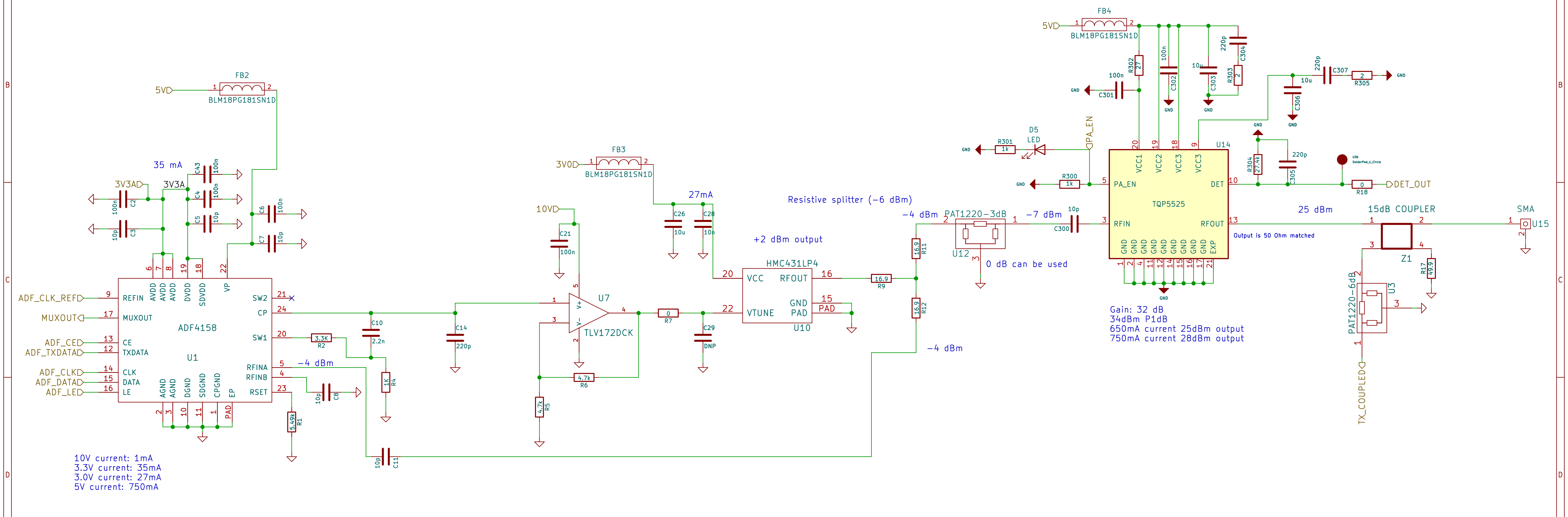
FPGA FIR Filter works perfectly.
Second stage for HPF
range compensation filer is removed

LT6232 has better noise performance as 1.1nV/√Hz
500 10K has less noise contribution

Sheet: /opamp_amp/
File: opamp_amp.kicad_sch

Title:

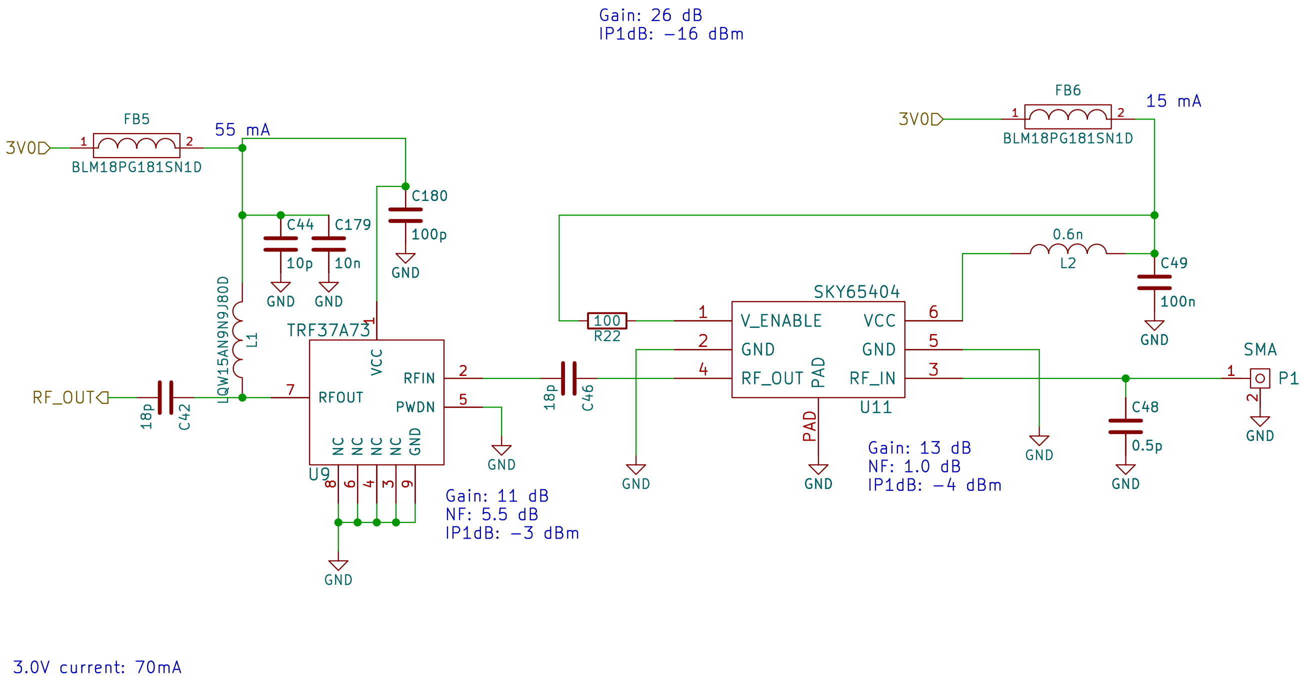
Size: A4 | Date: | Rev:
KiCad E.D.A. 9.0.7 | Id: 7/12



Sheet: /tx/
File: tx.kicad_sch

Title:

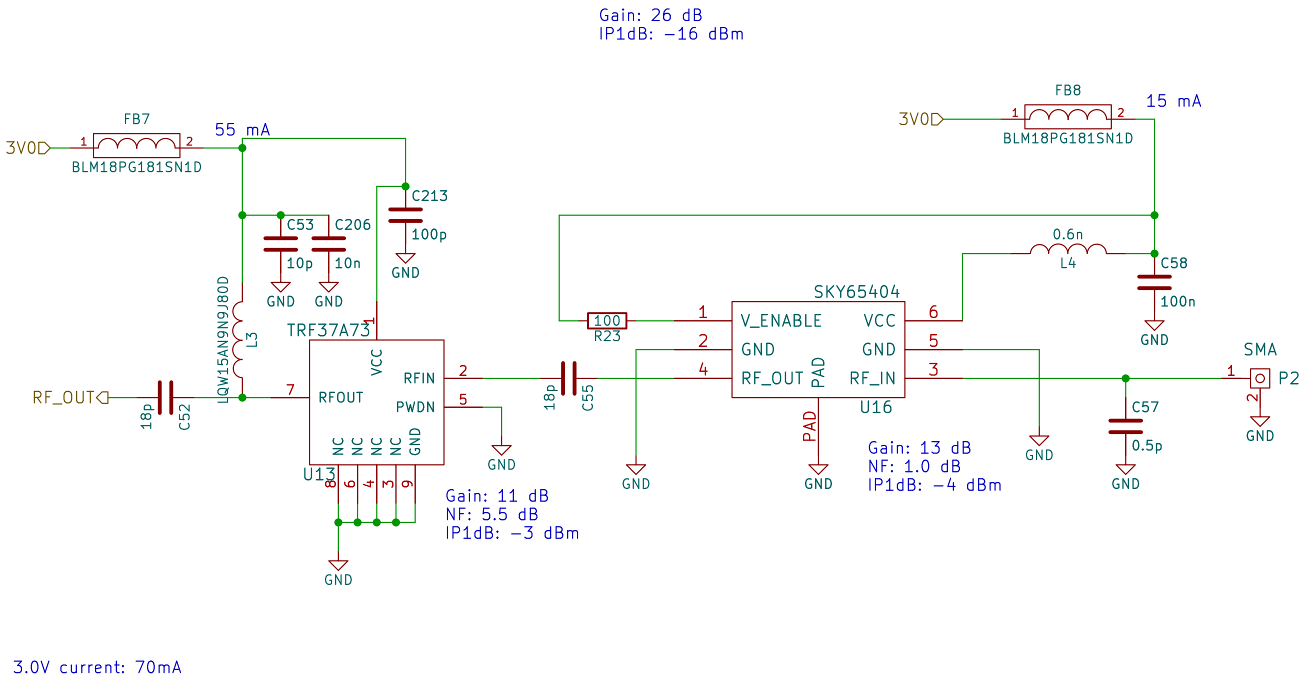
Size: A3 Date: Rev:
KiCad E.D.A. 9.0.7 Id: 8/12



Sheet: /rx1/
File: rx.kicad_sch

Title:

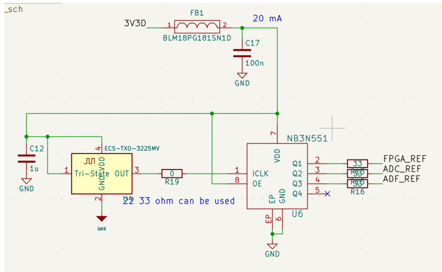
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 9/12



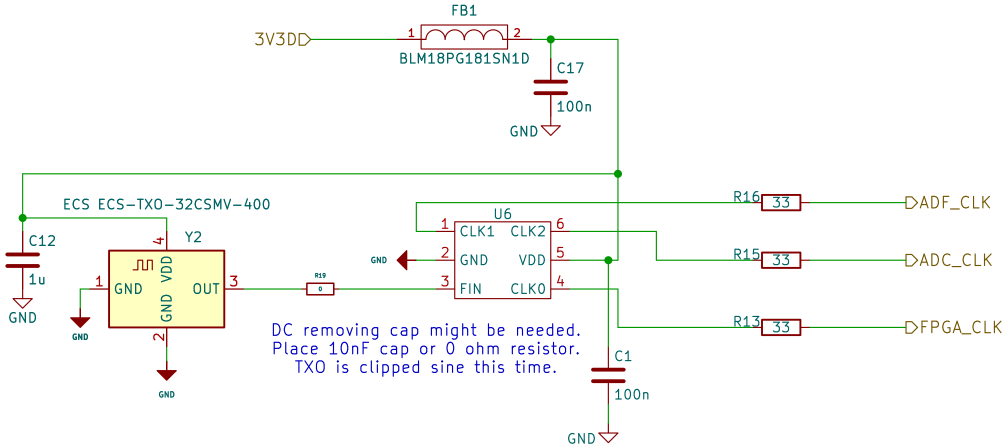
Sheet: /rx2/
File: rx.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.7		Id: 10/12



This is the prev design:
NB3N551 is obs.
Also this txo is generating pure sin.
ADC might need cmos like input not sin.
That could be the reason of duty stabilizer.



Sheet: /clock/
File: clock.kicad_sch

Title:

Size: A4	Date:	Rev:
KiCad E.D.A. 9.0.7		Id: 11/12